Virtual E. coli K-12

v0.1 · tap to fold

A slab cut through the middle of one cell, with every molecule drawn to scale. Drag to rotate, scroll or pinch to zoom, and click a molecule to identify it.

Slab thickness 40 nm

Re-pack

42,463 molecules placed in a 40 nm slab · 2,372 didn't fit (mostly the crowded outer membrane) · 90,150 nm of DNA · packed in 0.2 s · seed 2026

Layers

- [x] Lipid bilayers
- [x] LPS (core)
- [x] Peptidoglycan
- [x] Outer membrane
- [x] Periplasm
- [x] Inner membrane
- [x] Ribosomes & tRNA
- [x] Protein synthesis & folding
- [x] Metabolic enzymes
- [x] Rest of proteome
- [x] Chromosomal DNA
- [x] Nucleoid proteins
- [x] Polymerases & topos
- [x] Whole-cell outline (1 × 2 µm)

Find a molecule

Gene or name, e.g. groL, ribosome, porin

▸ All molecules in this model (158)

Molecule list, PDB sources and colours follow D. S.

Click a molecule to identify it

Goodsell, *Integrative Illustration of Escherichia coli* (RCSB PDB, 2022, doi:10.2210/rcsb_pdb/goodsell-gallery-028). Copy numbers per cell are rough placeholders to be replaced with measured values (e.g. Li et al., *Cell* 2014). Shapes are coarse sphere clusters sized from molecular mass, not real PDB shapes yet. Packing is an autoPACK-style random sequential fill (largest first, compartment-aware) computed in your browser; the cytoplasm is filled to ~30% w/v macromolecules. K-12 makes no O-antigen, so only the LPS core is drawn.

